Vision Transformer Depth

CIFAR-10 comparison of two and four encoder blocks

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Checkpoint results

Depth	Best epoch	Best validation accuracy	Test accuracy
2	20 / 20	68.48%	67.20%
4	19 / 20	73.48%	72.60%

Each row uses the earliest checkpoint with the highest validation accuracy. The selected checkpoint was evaluated once on all 10,000 test images.

Learning curves

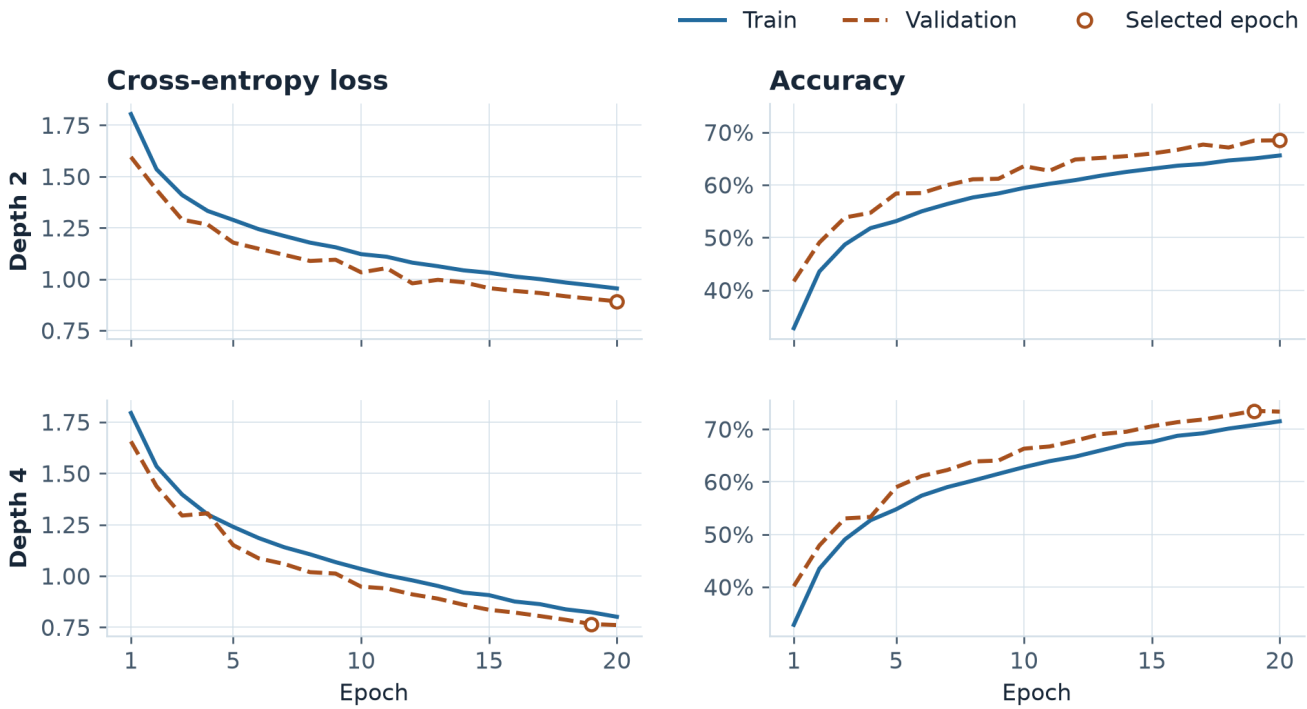


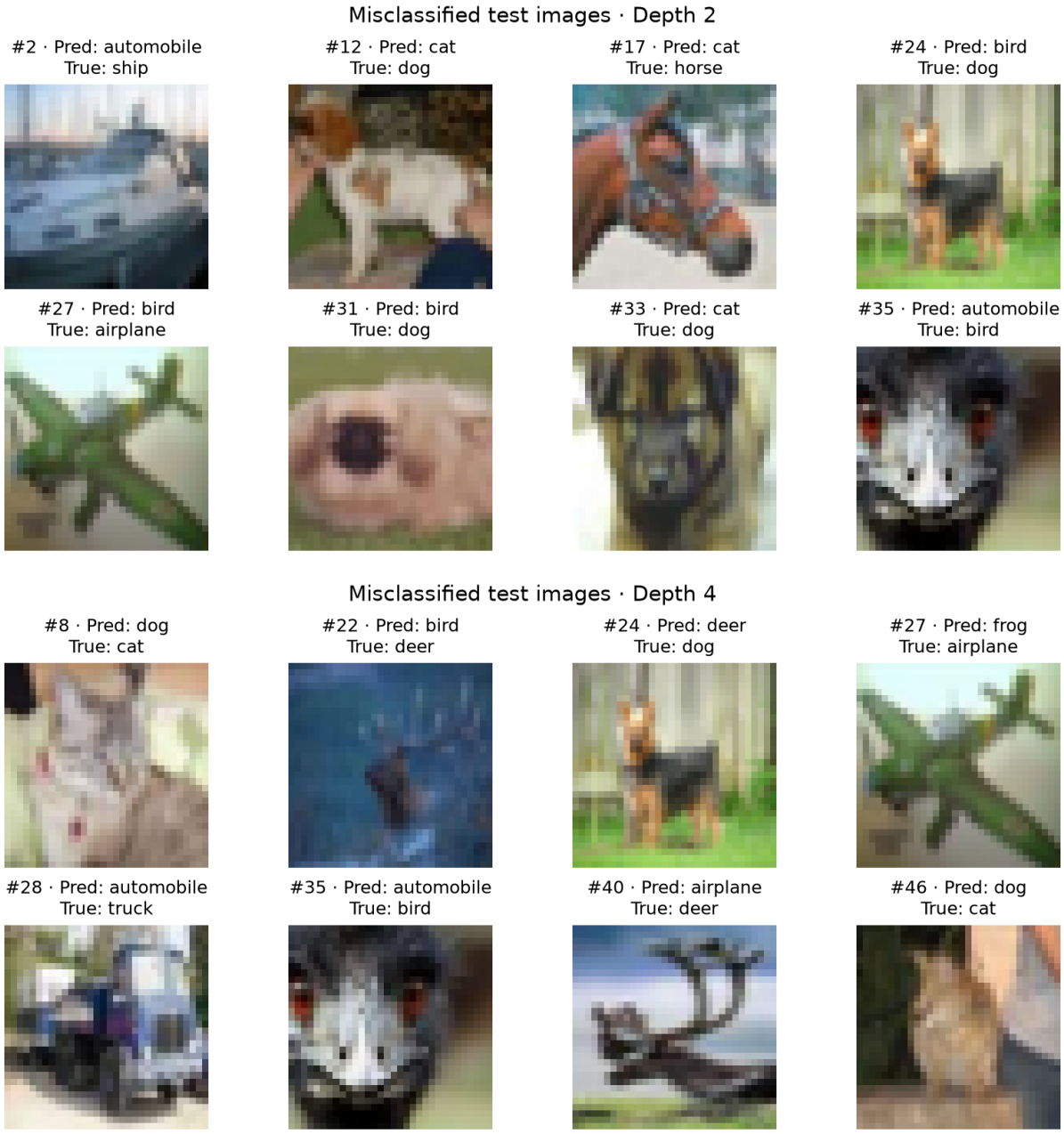
Figure 1. Training and validation loss and accuracy by epoch. Open circles mark the checkpoint selected by validation accuracy.

Experimental setup

Both runs used 45,000 training and 5,000 validation images with split seed 42. Inputs stayed at 32 x 32 pixels. Shared settings included 4 x 4 patches, 128-dimensional embeddings, four attention heads, AdamW (learning rate 0.0003, weight decay 0.05), and batch size 128. Both ran for 20 epochs on an RTX 5050 Laptop GPU.

Misclassified test images

Eight errors per depth, collected during the test evaluation. Numbers are zero-based CIFAR-10 test indices.



Depth comparison

Depth 4 improved best validation accuracy by 5.00 percentage points and test accuracy by 5.40 points. It reached 60% validation accuracy at epoch 6, two epochs earlier than depth 2. Depth 4 peaked at epoch 19 and dipped by 0.14 points in the final epoch, while depth 2 peaked at epoch 20. Validation loss ended at its lowest value for both depths, so the curves show little evidence of sustained overfitting within 20 epochs.